# Variation of Parameters

- **Motivation.** For $y'' + P(x)y' + Q(x)y = R(x)$, we need a particular solution $y_p$.
- The method of **undetermined coefficients** only works for special forms of $R(x)$.
- **Variation of parameters** works for *any* continuous $R(x)$: $\tan x, \sec x, \ln x, 1/x$, etc.
- It only requires knowing the complementary solution $y_c = c_1 y_1 + c_2 y_2$.
- The key idea is to replace the constants $c_1, c_2$ by unknown functions $u_1(x), u_2(x)$.
- We want the particular solution as $y_p = u_1(x)y_1(x) + u_2(x)y_2(x)$.
- This is why the method is called "variation of parameters."



# Variation of Parameters

- **Setup.** Let $y_1, y_2$ be a fundamental set of solutions of the corresponding H. Equation: $y'' + P(x)y' + Q(x)y = 0$.
- Assume $y_p = u_1 y_1 + u_2 y_2$ for unknown functions $u_1(x), u_2(x)$.
- Differentiating w. r. t. x gives us

  $y_p' = u_1' y_1 + u_1 y_1' + u_2' y_2 + u_2 y_2'$.
- To avoid second derivatives of $u_1, u_2$ appearing later, **impose the constraint**:

$$u_1' y_1 + u_2' y_2 = 0. \tag{5}$$

- Thus, we obtain another equation as

$$y_p' = u_1 y_1' + u_2 y_2'.$$

- Differentiating the above expression w. r. t $x$ again,

  $y_p'' = u_1' y_1' + u_1 y_1'' + u_2' y_2' + u_2 y_2''$.



# Variation of Parameters

- Substitute $y_p, y_p', y_p''$ into $y'' + P(x)y' + Q(x)y = R(x)$:
  $u_1(y_1'' + Py_1' + Qy_1) + u_2(y_2'' + Py_2' + Qy_2) + u_1'y_1' + u_2'y_2' = R(x)$
- Since $y_1, y_2$ solve the homogeneous equation, the bracketed terms vanish, and we obtain

$$u_1'y_1' + u_2'y_2' = R(x). \tag{6}$$

- Thus, (5) and (6) gives a $2 \times 2$ linear system in $u_1', u_2'$.
- The determinant of this system is the **Wronskian** $W(y_1, y_2) = y_1y_2' - y_2y_1' \neq 0$.
- By Cramer's rule:

$$u_1' = \frac{-y_2R(x)}{W}, \qquad u_2' = \frac{y_1R(x)}{W}.$$

- Integrate to obtain $u_1(x)$ and $u_2(x)$.



# Variation of Parameters

- **Thus, the Final formulas are**

$$u_1(x) = -\int \frac{y_2(x)R(x)}{W(x)}\,dx$$

$$u_2(x) = \int \frac{y_1(x)R(x)}{W(x)}\,dx$$

- Particular solution:

$$\boxed{y_p = y_1 \int \frac{-y_2 R}{W}\,dx + y_2 \int \frac{y_1 R}{W}\,dx}$$

- General solution of $y'' + P(x)y' + Q(x)y = R(x)$ is

$$y = c_1 y_1 + c_2 y_2 + y_p$$

- No arbitrary constants are kept in $u_1, u_2$ (they are absorbed into $y_p$).



# Example (VP)

- **Q.** Solve $y'' + y = \tan x,\ -\frac{\pi}{2} < x < \frac{\pi}{2}$.
- Homogeneous solutions: $y_1 = \cos x,\ y_2 = \sin x$.
- Wronskian: $W = \cos x(\cos x) - \sin x(-\sin x) = 1$.

  $u_1' = -y_2 \tan x = -\sin x \tan x = -\dfrac{\sin^2 x}{\cos x}$

  $u_2' = y_1 \tan x = \cos x \tan x = \sin x$
- Integrate: $u_2 = -\cos x$.
- For $u_1$: write $\dfrac{\sin^2 x}{\cos x} = \sec x - \cos x$.

  $u_1 = -\displaystyle\int (\sec x - \cos x)\, dx = -\ln|\sec x + \tan x| + \sin x$
- Compute $y_p = y_1 u_1 + y_2 u_2$:

  $y_p = \cos x\big(-\ln|\sec x + \tan x| + \sin x\big) + \sin x(-\cos x)$

  $= -\cos x \ln|\sec x + \tan x| + \sin x \cos x - \sin x \cos x$

  PS is $\boxed{y_p = -\cos x \ln|\sec x + \tan x|}$

  GS is $\boxed{y = c_1 \cos x + c_2 \sin x - \cos x \ln|\sec x + \tan x|}$.



# Example (VP)

- **Q.** Solve $y'' - 2y' + y = \dfrac{e^x}{x}$, $x > 0$.
- Characteristic equation: $r^2 - 2r + 1 = 0 \;\Rightarrow\; r = 1$ (repeated).
- Homogeneous solutions: $y_1 = e^x$, $y_2 = xe^x$.
- Wronskian: $W = e^x(e^x + xe^x) - xe^x(e^x) = e^{2x}$.

  $$u_1' = \frac{-y_2 R}{W} = \frac{-xe^x \cdot (e^x/x)}{e^{2x}} = -1 \;\Rightarrow\; u_1 = -x$$

  $$u_2' = \frac{y_1 R}{W} = \frac{e^x \cdot (e^x/x)}{e^{2x}} = \frac{1}{x} \;\Rightarrow\; u_2 = \ln x.$$
- Compute $y_p = y_1 u_1 + y_2 u_2$:

  $y_p = e^x(-x) + xe^x \ln x = xe^x(\ln x - 1)$

  $\boxed{y_p = xe^x(\ln x - 1)}$
- General solution:

  $\boxed{y = c_1 e^x + c_2 xe^x + xe^x(\ln x - 1)}$
- Note: $\dfrac{e^x}{x}$ is not of the standard undetermined-coefficients form, so variation of parameters is essential here.



# Example (VP)

- **Q.** Solve $y'' + 4y = \sec 2x$.
- Homogeneous solutions: $y_1 = \cos 2x$, $y_2 = \sin 2x$.
- Wronskian: $W = \cos 2x(2\cos 2x) - \sin 2x(-2\sin 2x) = 2$.

$$u_1' = \frac{-y_2 R}{W} = \frac{-\sin 2x \sec 2x}{2} = -\frac{1}{2}\tan 2x$$

$$u_2' = \frac{y_1 R}{W} = \frac{\cos 2x \sec 2x}{2} = \frac{1}{2}$$

- Integrate: $u_2 = \frac{x}{2}$, and $u_1 = -\frac{1}{2}\int \tan 2x\, dx = \frac{1}{4}\ln|\cos 2x|$.
- Then the PS $y_p = y_1 u_1 + y_2 u_2$ is

$$y_p = \cos 2x \cdot \frac{1}{4}\ln|\cos 2x| + \sin 2x \cdot \frac{x}{2}$$

$$\boxed{y_p = \frac{1}{4}\cos 2x \ln|\cos 2x| + \frac{x}{2}\sin 2x}$$

- General solution is

$$\boxed{y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{4}\cos 2x \ln|\cos 2x| + \frac{x}{2}\sin 2x}$$



# Higher Order Homogeneous ODEs

- An $n$-th order linear homogeneous ODE with constant coefficients has the form
  $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0, \quad a_n \neq 0.$
- We need the solutions of the form $y = e^{rx}$.
- Substituting gives $a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = 0$.
- This is called the **auxiliary** or **characteristic equation**.
- It is a degree-$n$ polynomial in $r$, so it has exactly $n$ roots (real or complex, with multiplicity).
- Each root determines one basis solution.
- If we take all these solutions together, they form the solution space of the DE.
- Dimension of the solution space is $n$.



# Higher Order Homogeneous ODEs

- **Case 1: Distinct Real Roots.**
- If $r_1, r_2, \ldots, r_n$ are all real and distinct,
- then the corresponding basis solutions are $e^{r_1 x}, e^{r_2 x}, \ldots, e^{r_n x}$.
- These are linearly independent (the generalized Wronskian is nonzero).
- General solution:

  $y = c_1 e^{r_1 x} + c_2 e^{r_2 x} + \cdots + c_n e^{r_n x}$
- This directly generalizes the 2nd order distinct-roots case.

